## LAB EXPERIMENTS 1 TO 6

1. Scenario: You are working on a project that involves analyzing student performance data for a class of 32 students. The data is stored in a NumPy array named student_scores, where each row represents a student and each column represents a different subject. The subjects are arranged in the following order: Math, Science, English, and History. Your task is to calculate the average score for each subject and identify the subject with the highest average score.
Question: How would you use NumPy arrays to calculate the average score for each subject and determine the subject with the highest average score? Assume 4x4 matrix that stores marks of each student in given order.
CODE:

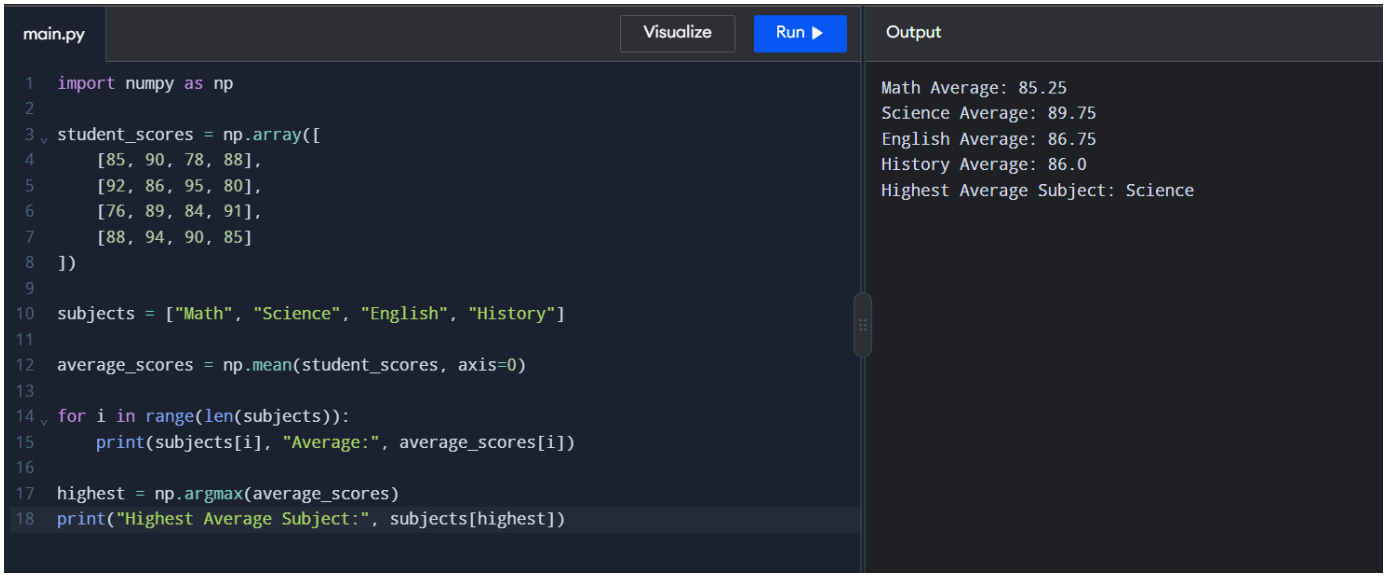
```
import numpy as np
student_scores = np.array([
    [85, 90, 78, 88],
    [92, 86, 95, 80],
    [76, 89, 84, 91],
    [88, 94, 90, 85]
])

subjects = ["Math", "Science", "English", "History"]

average_scores = np.mean(student_scores, axis=0)

for i in range(len(subjects)):
    print(subjects[i], "Average:", average_scores[i])

highest = np.argmax(average_scores)
print("Highest Average Subject:", subjects[highest])
```



2. Scenario: You are a data analyst working for a company that sells products online. You have been tasked with analyzing the sales data for the past month. The data is stored in a NumPy array.
Question: How would you find the average price of all the products sold in the past month? Assume 3x3 matrix with each row representing the sales for a different product.
CODE:

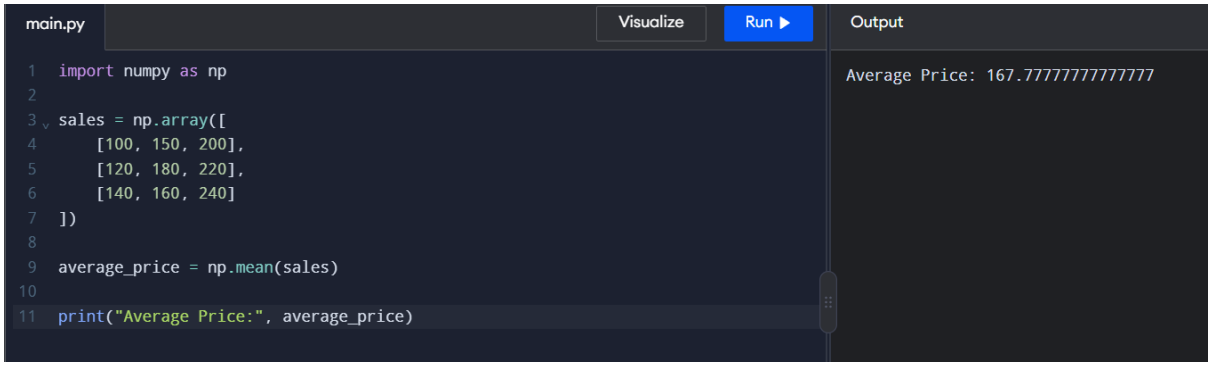
```
import numpy as np

sales = np.array([
    [100, 150, 200],
    [120, 180, 220],
    [140, 160, 240]
])

average_price = np.mean(sales)

print("Average Price:", average_price)
```

OUTPUT:




3. Scenario: You are working on a project that involves analyzing a dataset containing information about houses in a neighborhood. The dataset is stored in a CSV file, and you have imported it into a NumPy array named house_data. Each row of the array represents a house, and the columns contain various features such as the number of bedrooms, square footage, and sale price.
Question: Using NumPy arrays and operations, how would you find the average sale price of houses with more than four bedrooms in the neighborhood?
CODE:

```
import numpy as np

house_data = np.array([
    [3, 1500, 250000],
    [5, 2200, 400000],
    [6, 2800, 500000],
    [4, 1800, 300000],
```

```
    [5, 2400, 450000]
])

bedrooms = house_data[:, 0]
sale_prices = house_data[:, 2]

selected_prices = sale_prices[bedrooms > 4]
average_price = np.mean(selected_prices)

print("Average Sale Price:", average_price)
```




4. Scenario: You are working on a project that involves analyzing the sales performance of a company over the past four quarters. The quarterly sales data is stored in a NumPy array named sales_data, where each element represents the sales amount for a specific quarter. Your task is to calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter.
Question: Using NumPy arrays and arithmetic operations calculate the total sales for the year and determine the percentage increase in sales from the first quarter to the fourth quarter?
CODE:

```
import numpy as np

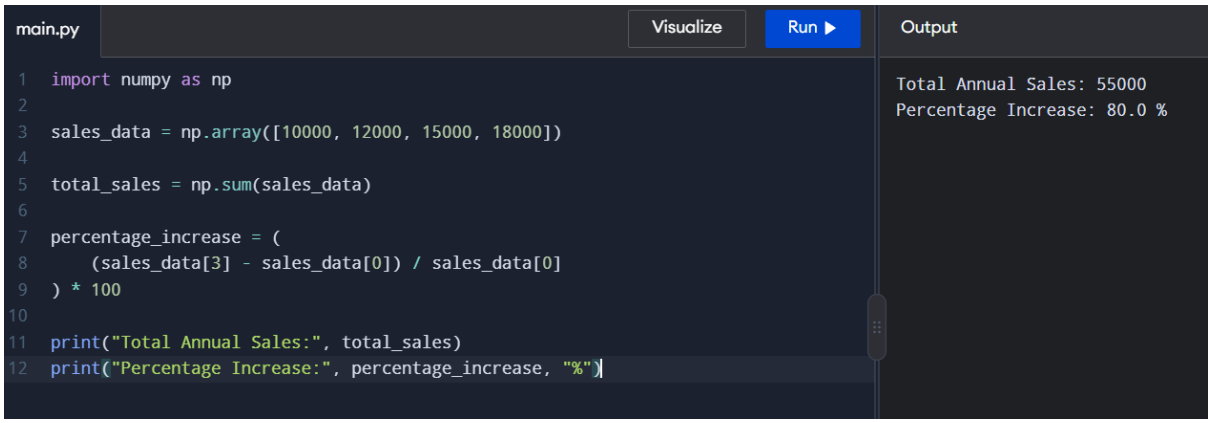
sales_data = np.array([10000, 12000, 15000, 18000])

total_sales = np.sum(sales_data)

percentage_increase = (
    (sales_data[3] - sales_data[0]) / sales_data[0]
) * 100

print("Total Annual Sales:", total_sales)
print("Percentage Increase:", percentage_increase, "%")
```

OUTPUT:




5. Scenario: You are a data analyst working for a car manufacturing company. As part of your analysis, you have a dataset containing information about the fuel efficiency of different car models. The dataset is stored in a NumPy array named fuel_efficiency, where each element represents the fuel efficiency (in miles per gallon) of a specific car model. Your task is to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models.
Question: How would you use NumPy arrays and arithmetic operations to calculate the average fuel efficiency and determine the percentage improvement in fuel efficiency between two car models?

CODE:

```
import numpy as np
fuel_efficiency = np.array([25, 30, 35, 40, 45])

average = np.mean(fuel_efficiency)

car1 = fuel_efficiency[0]
car2 = fuel_efficiency[1]

improvement = ((car2 - car1) / car1) * 100

print("Average Fuel Efficiency:", average, "mpg")
print("Percentage Improvement:", improvement, "%")
```

OUTPUT:

main.py

```
import numpy as np

fuel_efficiency = np.array([25, 30, 35, 40, 45])

average = np.mean(fuel_efficiency)

car1 = fuel_efficiency[0]
car2 = fuel_efficiency[1]

improvement = ((car2 - car1) / car1) * 100

print("Average Fuel Efficiency:", average, "mpg")
print("Percentage Improvement:", improvement, "%")
```

Output

```
Average Fuel Efficiency: 35.0 mpg
Percentage Improvement: 20.0 %
```

6. Scenario: You are a cashier at a grocery store and need to calculate the total cost of a customer's purchase, including applicable discounts and taxes. You have the item prices and quantities in separate lists, and the discount and tax rates are given as percentages. Your task is to calculate the total cost for the customer.
Question: Use arithmetic operations to calculate the total cost of a customer's purchase, including discounts and taxes, given the item prices, quantities, discount rate, and tax rate?

CODE:

```
prices = [100, 200, 300]
quantities = [2, 1, 3]

discount_rate = 10
tax_rate = 5
total = 0
for i in range(len(prices)):
    total += prices[i] * quantities[i]
discount = total * discount_rate / 100
discounted_total = total - discount
tax = discounted_total * tax_rate / 100
final_cost = discounted_total + tax
print("Total Cost:", total)
print("Discount:", discount)
print("Tax:", tax)
print("Final Cost:", final_cost)
```

OUTPUT:

main.py

```
prices = [100, 200, 300]
quantities = [2, 1, 3]

discount_rate = 10
tax_rate = 5

total = 0

for i in range(len(prices)):
    total += prices[i] * quantities[i]

discount = total * discount_rate / 100
discounted_total = total - discount

tax = discounted_total * tax_rate / 100
final_cost = discounted_total + tax

print("Total Cost:", total)
print("Discount:", discount)
print("Tax:", tax)
print("Final Cost:", final_cost)
```

Output

```
Total Cost: 1300
Discount: 130.0
Tax: 58.5
Final Cost: 1228.5
```